

ICU PREDICTION

Capstone 2 Final Technical Progress Presentation

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RECAP OF BALANCING STRATS

XGBoost Baseline Candidate (No Class Balancing)

XGBoost Classification Report

Metric	Precision	Recall	F1-Score
Class 0	0.90	0.98	0.94
Class 1	0.58	0.19	0.28
Macro Avg	0.74	0.58	0.61

ROC-AUC: 0.78
PRC-AUC: 0.39

Stacked Ensemble Baseline Candidate (With Class Balancing)

Stacked Ensemble Classification Report

Metric	Precision	Recall	F1-Score
Class 0	0.96	0.78	0.86
Class 1	0.32	0.78	0.46
Macro Avg	0.64	0.78	0.66

ROC-AUC: 0.81
PRC-AUC: 0.40

CHANGES SINCE OUR WHITEBOARD DEFENSE

Finalizing Deployment Demo (Toggle Option)

- Added a second-opinion toggle, helping with robustness
- Single → dual-model setup

Insight & Iteration

- Talked with professionals at SMU
- Explored underlying reasons behind our model limitations

Model Optimization & Experimentation

- Researched optimization strategies
- Experimented with wide range of techniques to improve performance (resampling, tuning, calibration, etc.)

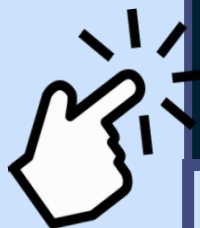
Performance Improvement

- Improved metrics beyond baseline performance!
- Evaluated different approaches to handle class imbalance



DEPLOYMENT DEMO

Balancing resource allocation and patient safety.



Get Second Opinion

ICU Risk Alert System

Patient ID	Bad Outcome Risk %	Risk Contributors	Recommended Course of Action
12345	3.4	Local SHAP Values	Standard monitoring
6789	87.42	Local SHAP Values	Immediate ICU Attention

DEPLOYMENT DEMO

Balancing resource allocation and patient safety.

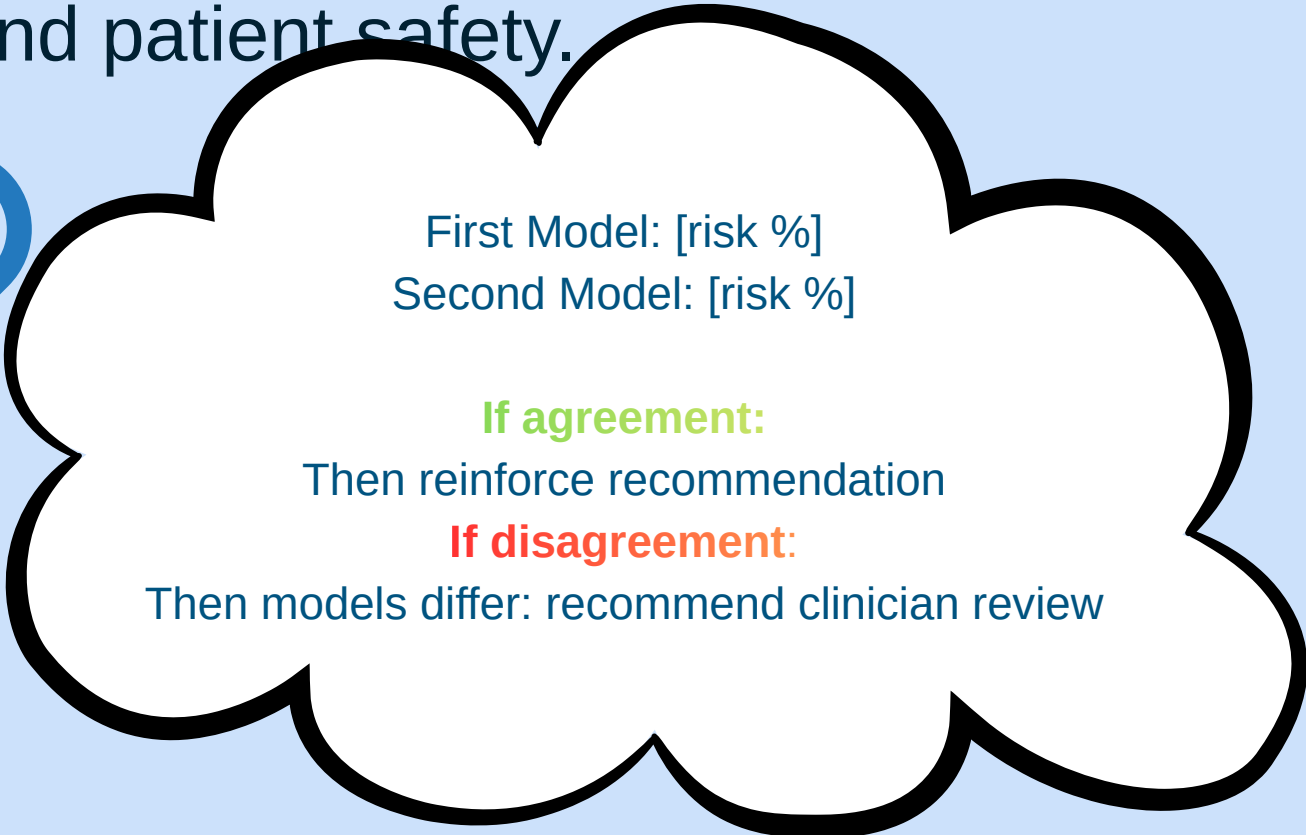


ICU Risk Alert System 2			
See First Opinion			
Patient ID	Bad Outcome Risk %	Risk Contributors	Recommended Course of Action
12345	3.4	Local SHAP Values	Standard monitoring
6789	87.42	Local SHAP Values	Standard Monitoring 

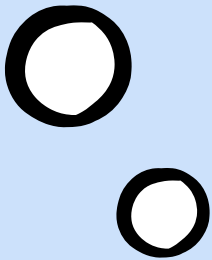


DEPLOYMENT DEMO

Balancing resource allocation and patient safety.



ICU Risk Alert System 2			
See First Opinion			
Patient ID	Bad Outcome Risk %	Risk Contributors	Recommended Course of Action
12345	3.4	Local SHAP Values	Standard monitoring
6789	87.42	Local SHAP Values	Standard Monitoring ⚠



OTHER STACKED ENSEMBLES IN THE ICU

A stacked ensemble deep learning model for predicting the intensive care unit patient mortality

Dimitrios Simopoulos^{1*} , Dimitrios Kosmidis² , George Anastassopoulos¹ , Lazaros Iliadis³ 

- Used same eICU dataset (full data)
- Stacked: 1 CatBoost + 2 Random Forests
- **Precision: 0.94**
- **Recall: 0.942**
- Used pre-calculated predictors for mortality/hospital death rate

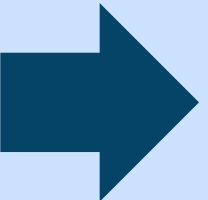
Enhancing predictions with a stacking ensemble model for ICU mortality risk in patients with sepsis-associated encephalopathy

[Xuhui Liu](#)^{1,2}, [Hao Niu](#)³, [Jiahua Peng](#)^{2,8}

- Used same eICU dataset (full data)
- **ROC-AUC: 0.671**
- **Precision: 0.808**
- **Recall: 0.808**
- Focused on subset of patients with Sepsis-associated encephalopathy (SAE)

TECHNIQUES EXPLORED

Category	Methods
Single Models	Logistic Regression, XGBoost, LightGBM, Random Forest, SVM, Naive Bayes, Multi-Layer Perceptron (MLP)
Ensemble Strategies	Stacking (LR + NB + RF + XGB → Meta LR), Stacking (SVM + XGB → Meta LR), Stacking (LR + XGB → Meta XGB)
Training Strategies (Applied to Base Learners Only)	Stratified Train/Test Split (70/30, 80/20), Stratified K-Fold Cross-Validation (5-fold), Out-of-Fold (OOF) Predictions for Stacking, Nested CV
Class Imbalance Handling (Done Inside Folds)	Random Oversampling, No Random Undersampling, SMOTE, Borderline-SMOTE, ADASYN, Class Weight Balancing, KMeansSmote, SMOTE-ENN
Threshold Optimization	Default Threshold (0.5), F1-score Optimization, Cost-sensitive Thresholding (FP vs FN weighting)
Calibration Techniques	Isotonic, Sigmoid, Quantile
Feature Engineering	ICU unit encoding, shock indicators, severity scores, lab ratios, missing value flags, removed noisy categorical features → 55 to 65 columns
Hyperparameter Tuning	RandomizedSearchCV, GridSearchCV

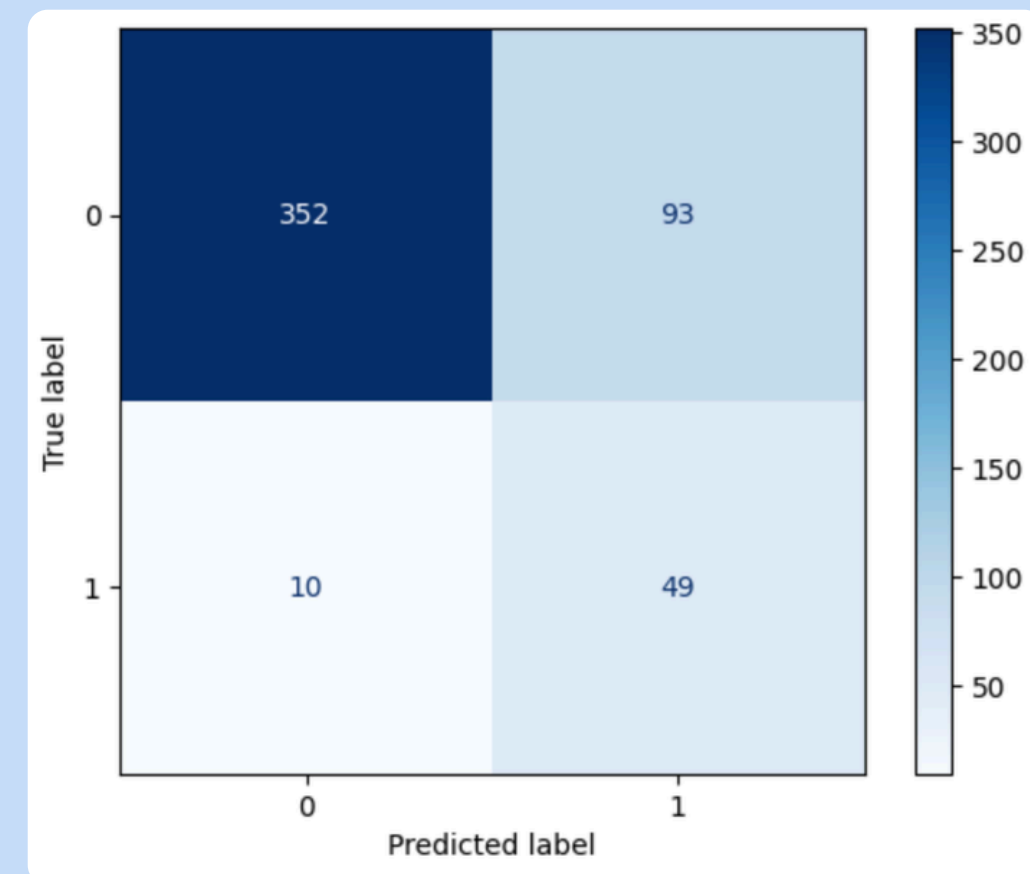


PRIMARY MODEL

- LR + XGB (Base) → XGB (Meta)
- Random Search
- Random Oversampling
- Class imbalance handled with class weighting
- 80/20 split with stratified K-Fold cross-validation with OOF used to train meta
- Probability calibration using Isotonic Regression
- Cost-based threshold tuning (5:1 emphasis on minimizing false negatives vs false positives)

ICU Model Performance Comparison

Metric	Baseline Stacked	Optimized (XGB Meta)	Improvement
Class 1 Precision	0.32	0.42	+0.10
Class 1 Recall	0.78	0.83	+0.05
Class 1 F1-Score	0.46	0.54	+0.08
ROC AUC	0.81	0.845	+0.035
PR AUC	0.40	0.45	+0.05



SUPPORTING MODEL



Refining the Original Stacked Model

Explored Techniques:

- Different splits (70/15/15 vs 80/20)
- SMOTE / resampling
- Multiple stacking variations



Selected Approach:

- 70/15/15 split
- Nested / staged validation workflow
- Calibration comparison
- Final calibrated stacked logistic meta-model
- Deployment threshold = 0.15

MODEL RESULTS

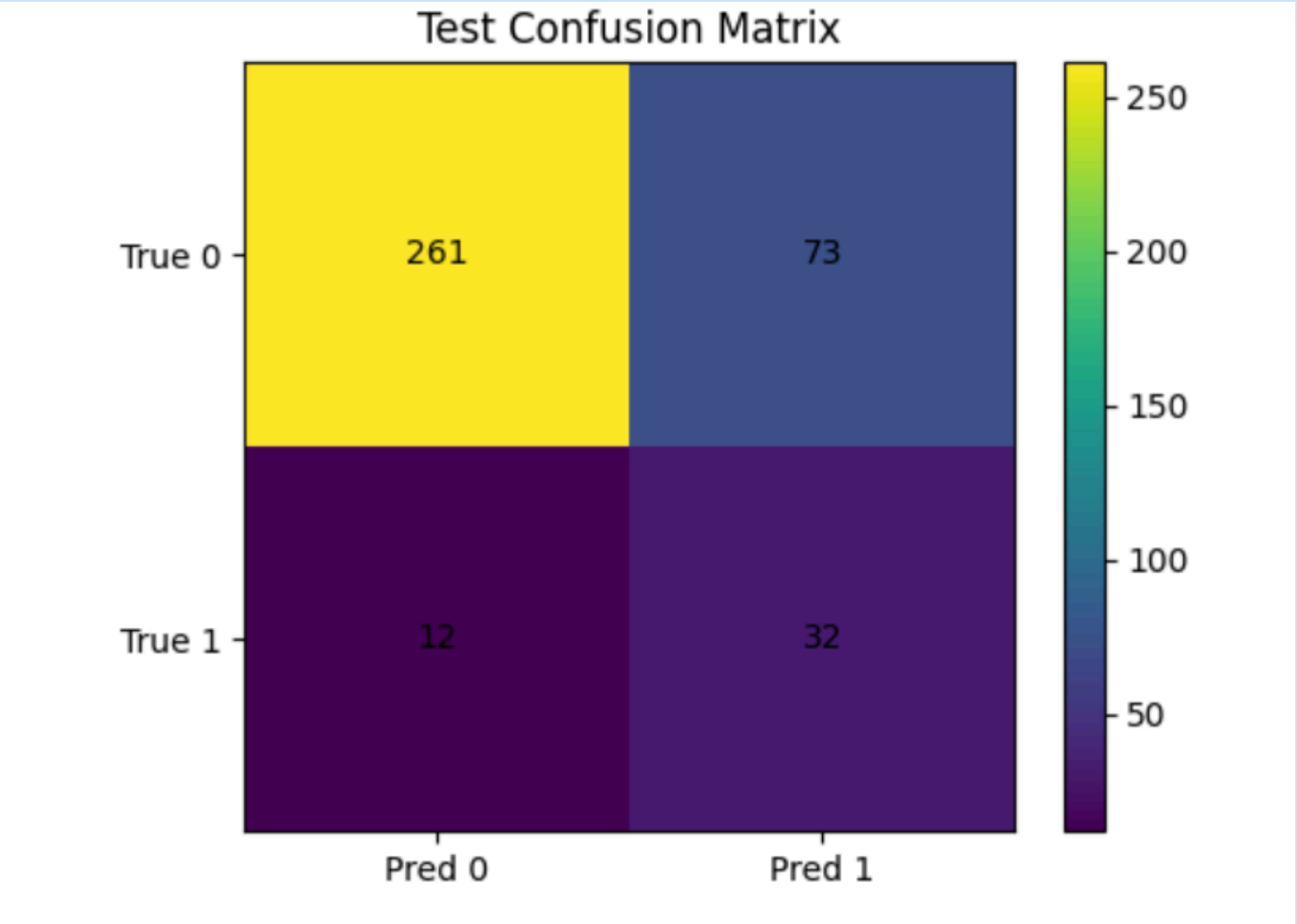
Test Final Metrics and Classification Report

Table 1: Final Test Metrics (with threshold = 0.15)

	Auroc	Auprc	F1	Brier	tn	fp	fn	tp
0	0.8256	0.3819	0.4295	0.087	261	73	12	32

Table 2: Test Classification Report

Metric	Precision	Recall	F1-Score	Support
Class 0	0.9560	0.7814	0.8600	334
Class 1	0.3048	0.7273	0.4295	44
Accuracy			0.7751	378
Macro Avg	0.6304	0.7544	0.6447	378
Weighted Avg	0.8802	0.7751	0.8099	378



- Base learners: LogReg, SVM, Random Forest, XGBoost
- Meta-learner: Logistic Regression

Role of This Model & Trade-offs

Model Performance Comparison

Metric	My Model	Optimized Model 
Recall	0.73	0.83
Precision	0.30	0.42
ROC	0.83	0.845
PRC	0.38	0.45
Calibration	Yes	Yes

Strengths:

- Maintains strong recall (0.73)
- Only 12 high-risk patients missed
- Calibrated probabilities (Brier = 0.087)

Trade-offs:

- Lower precision (~0.30)
- More false positives
- Slightly weaker ranking (PR AUC)

Why:

- Provides a second perspective
- More reliable probability estimates
- Useful for decision support + validation

WHY OUR CHANGES = MORE EXPLAINABILITY

1. Optimized Hyperparameters → More Trustworthy SHAP Values

2. Cleaner Stacking → Cleaner Attribution

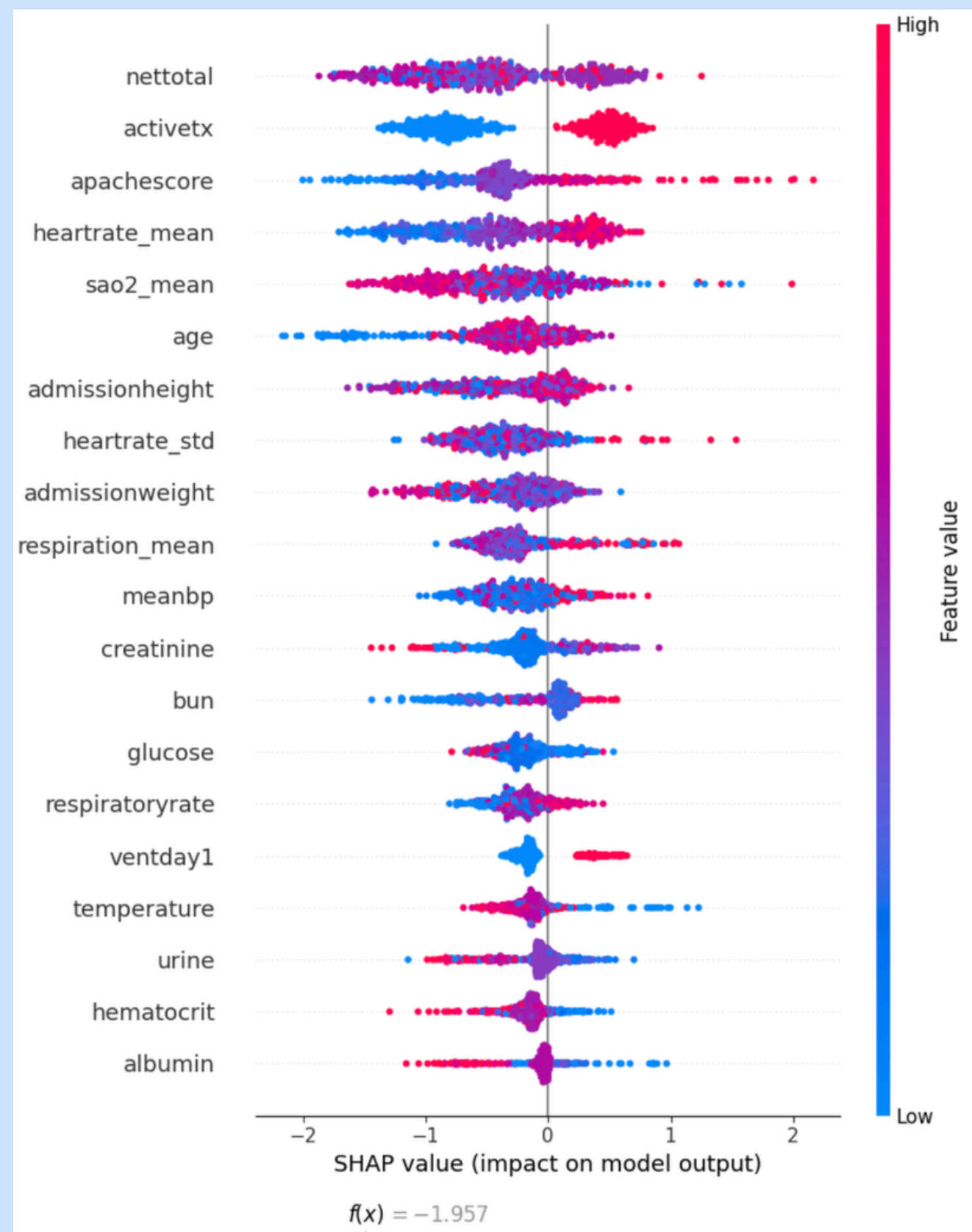
3. Deliberate Threshold Strategy → Explainability at the Decision Boundary

4. Better Calibration → SHAP Values in Probability Space

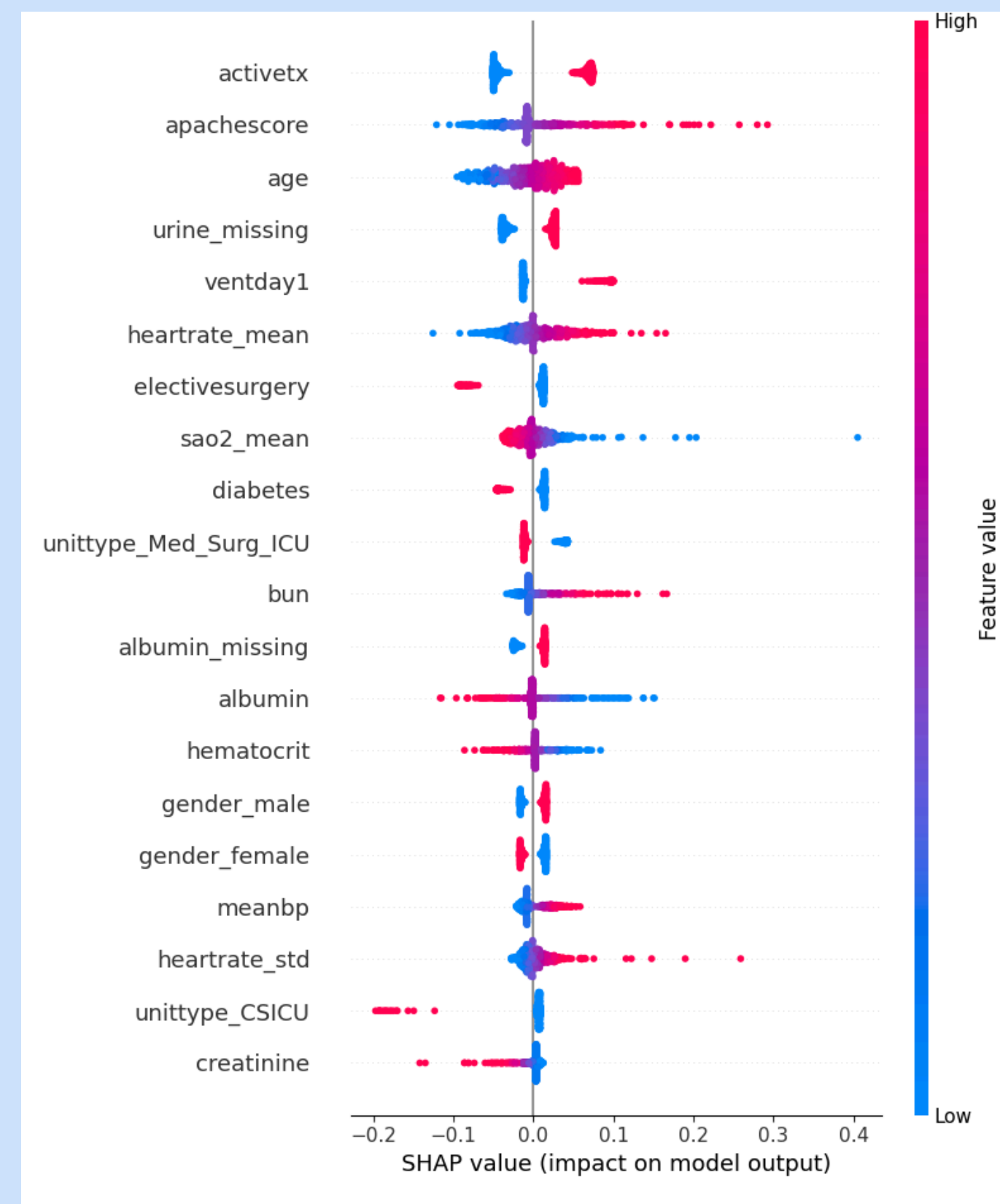


GLOBAL EXPLAINABILITY



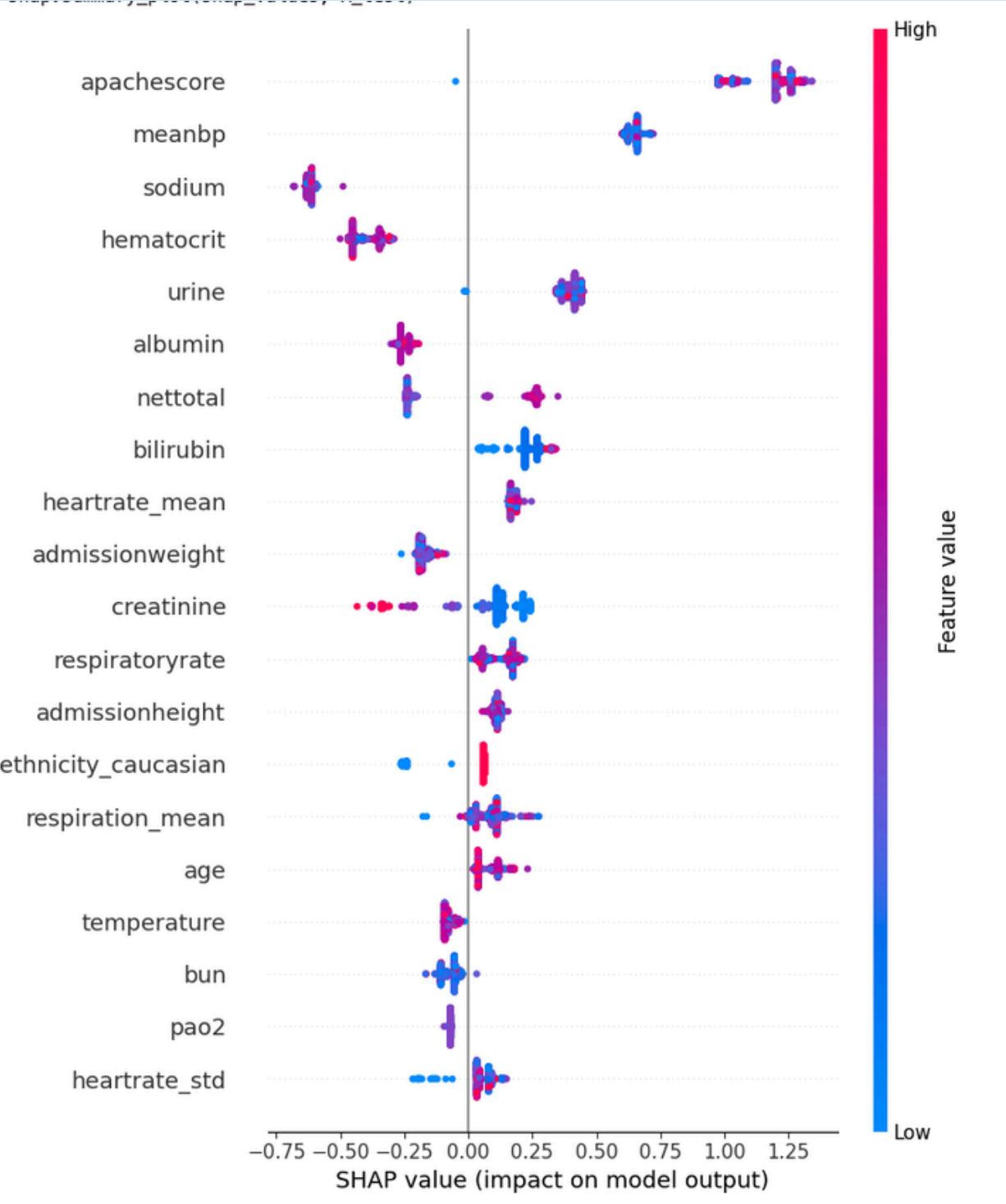


XGBOOST

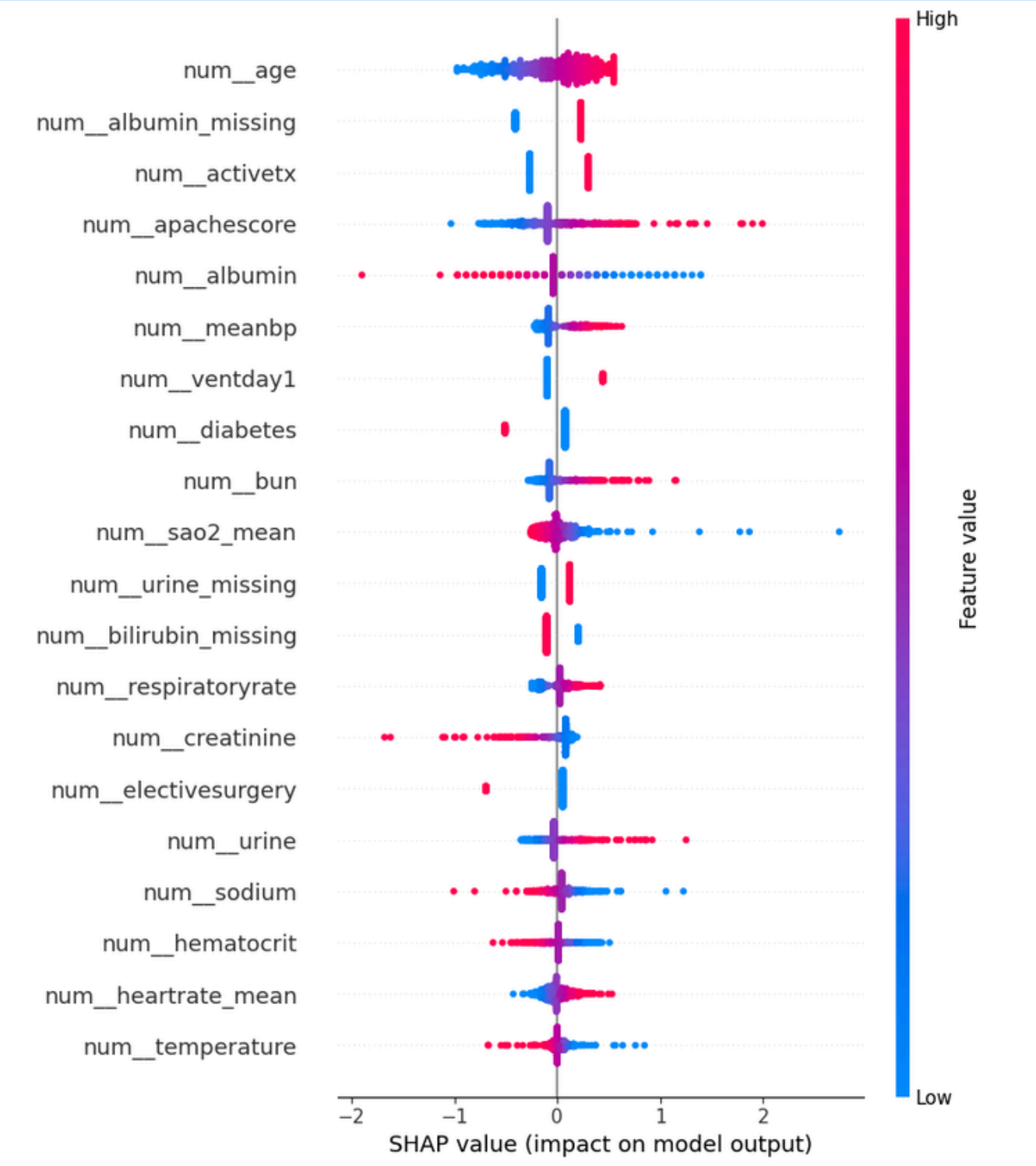


LR

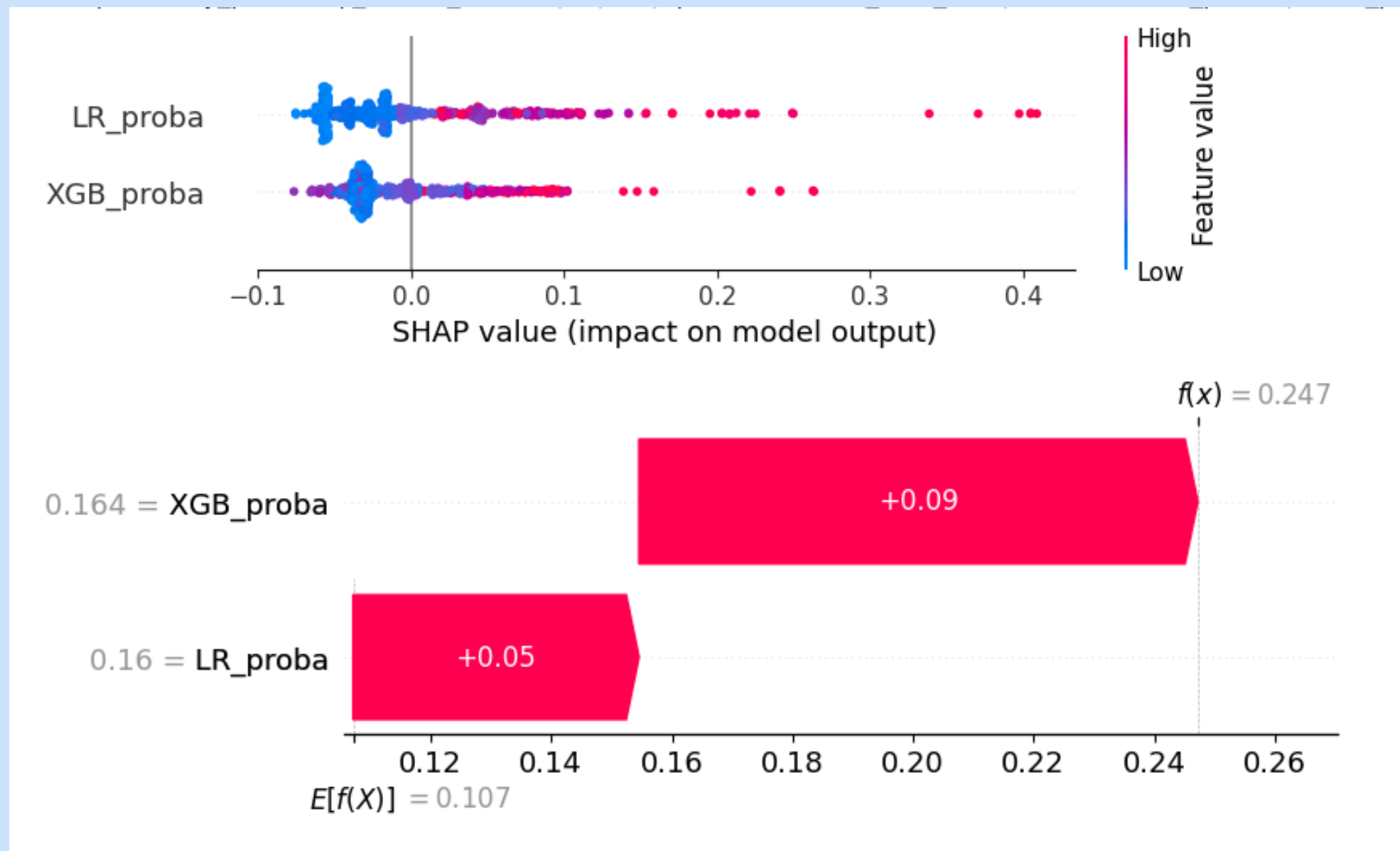
SUPPORTING MODEL



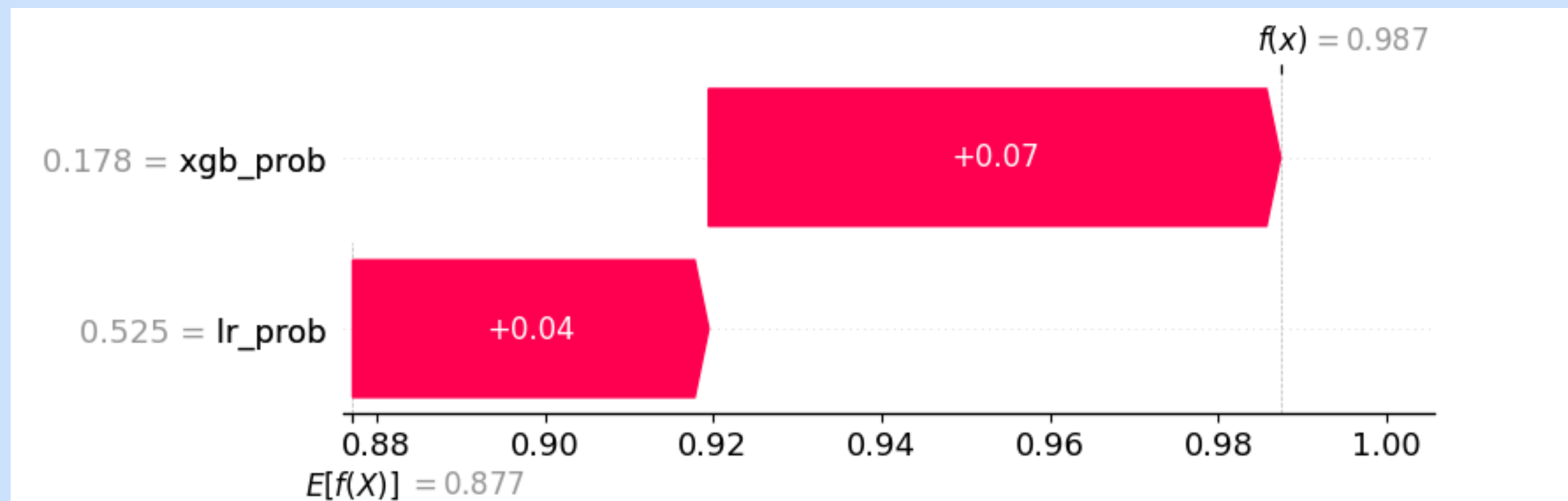
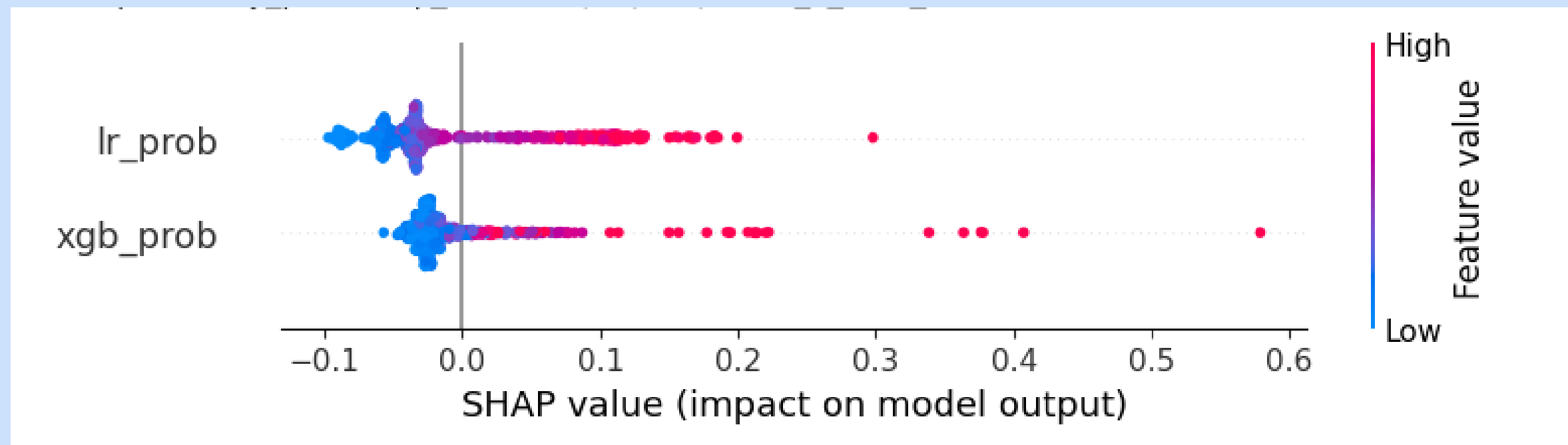
XGBOOST



LR



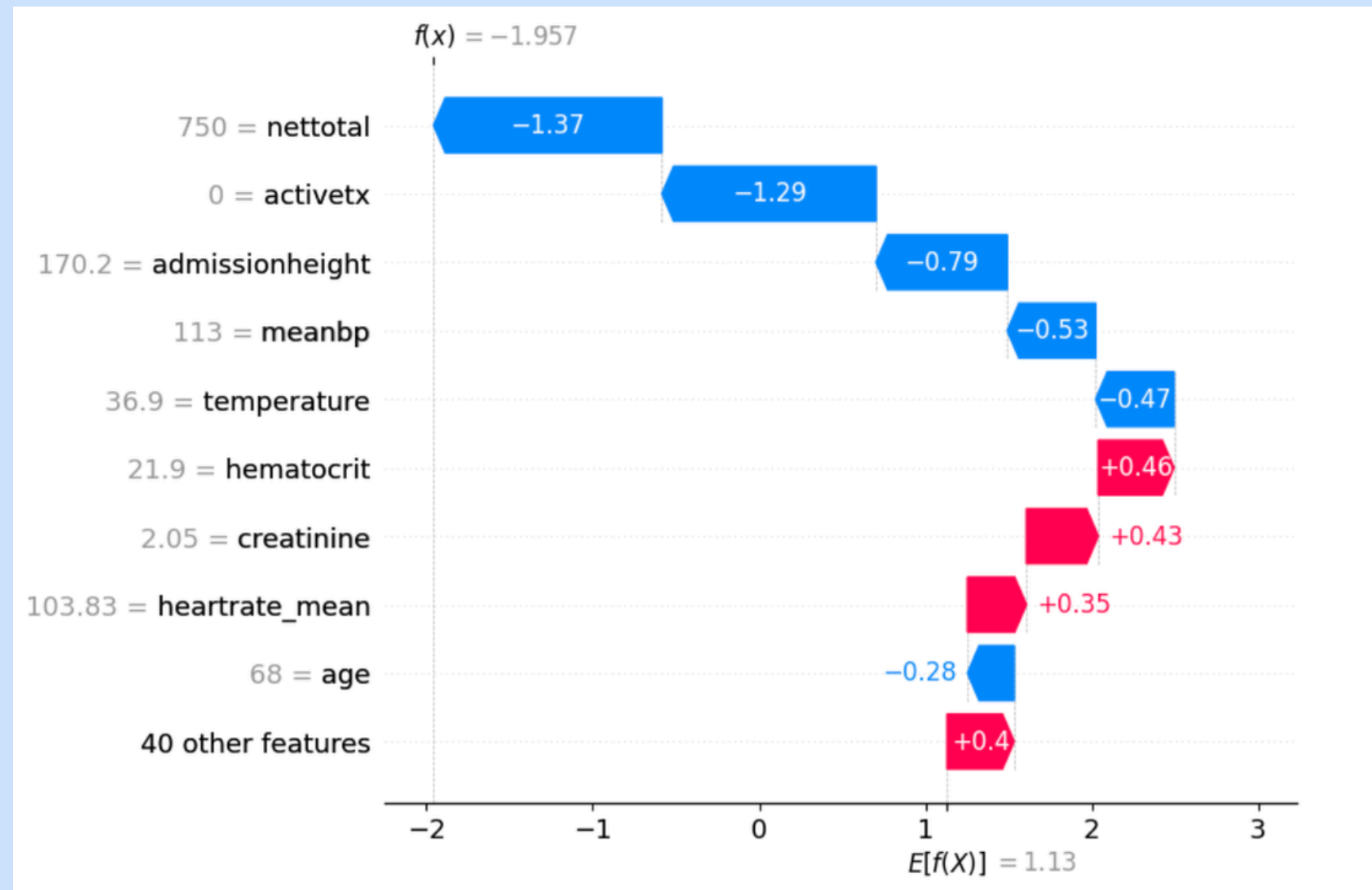
OPTIMIZED STACKED MODEL



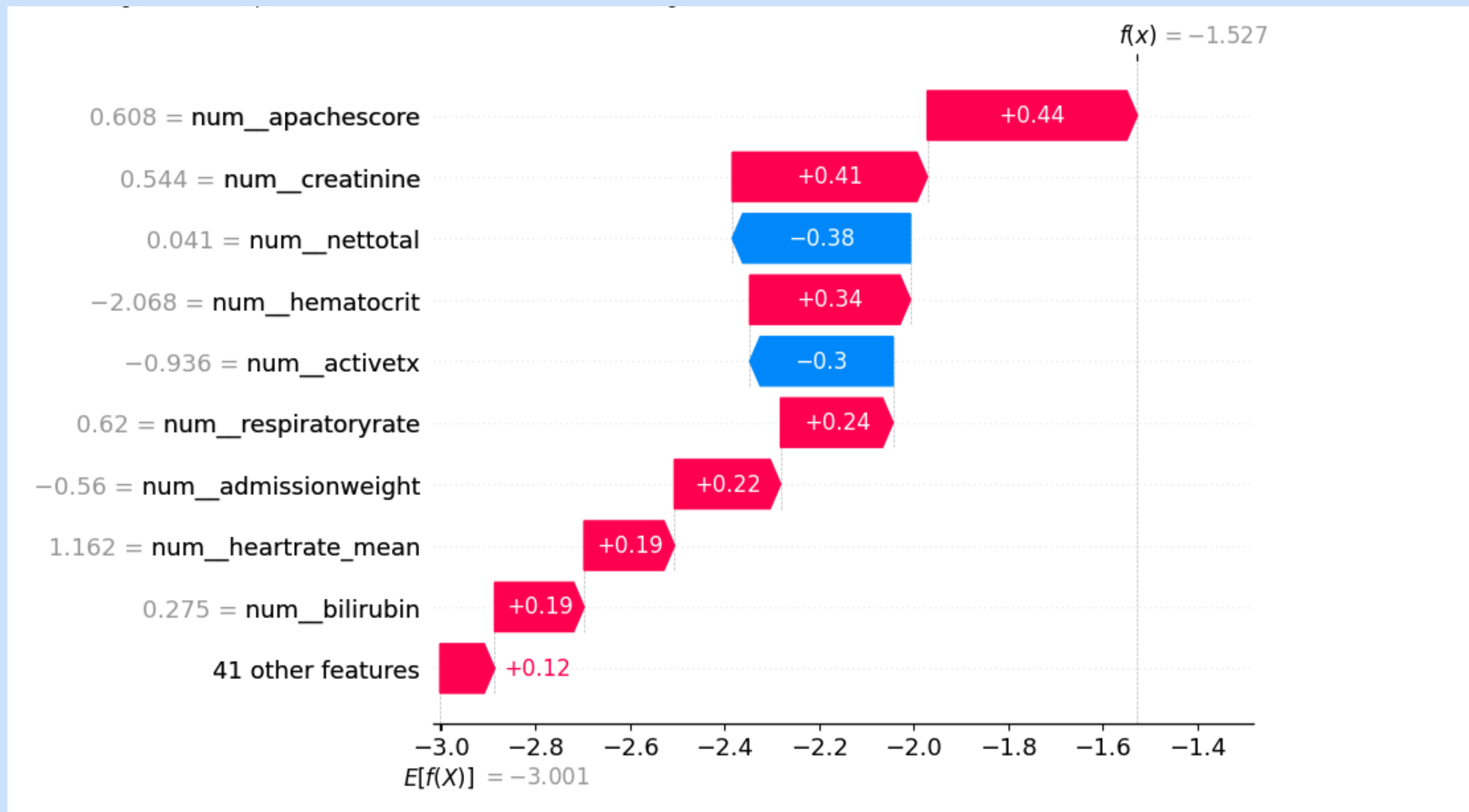
SUPPORTING MODEL

LOCAL EXPLAINABILITY +





OPTIMIZED STACK MODEL LOCAL SHAP

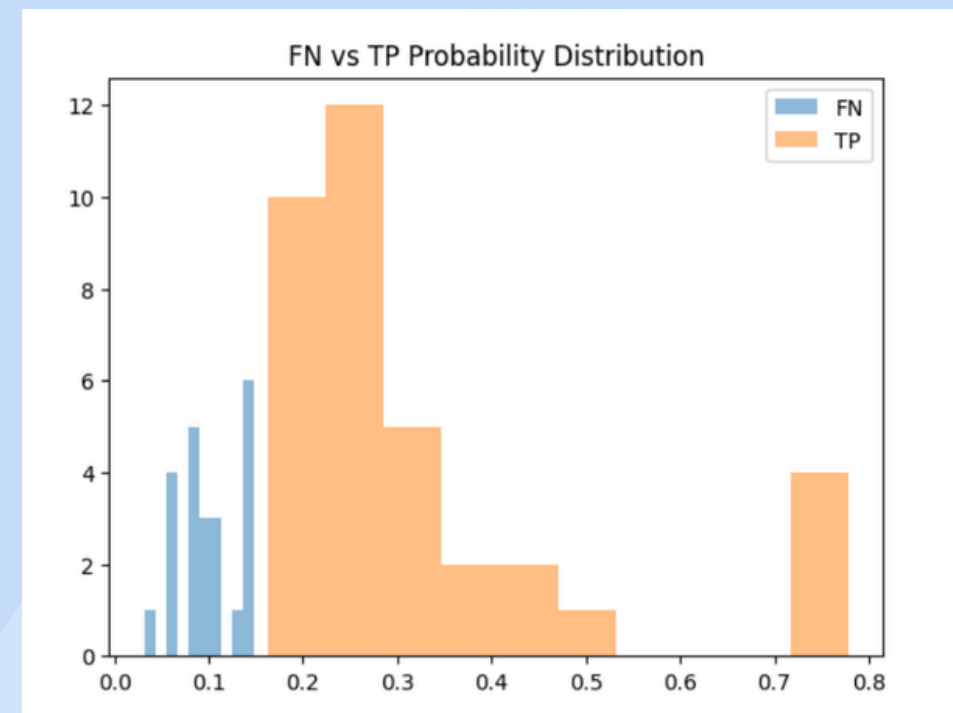


SUPPORTING MODEL LOCAL SHAP

+ERROR ANALYSIS

composite outcome - while mortality and readmission are bad outcomes, they both have different underlying drivers

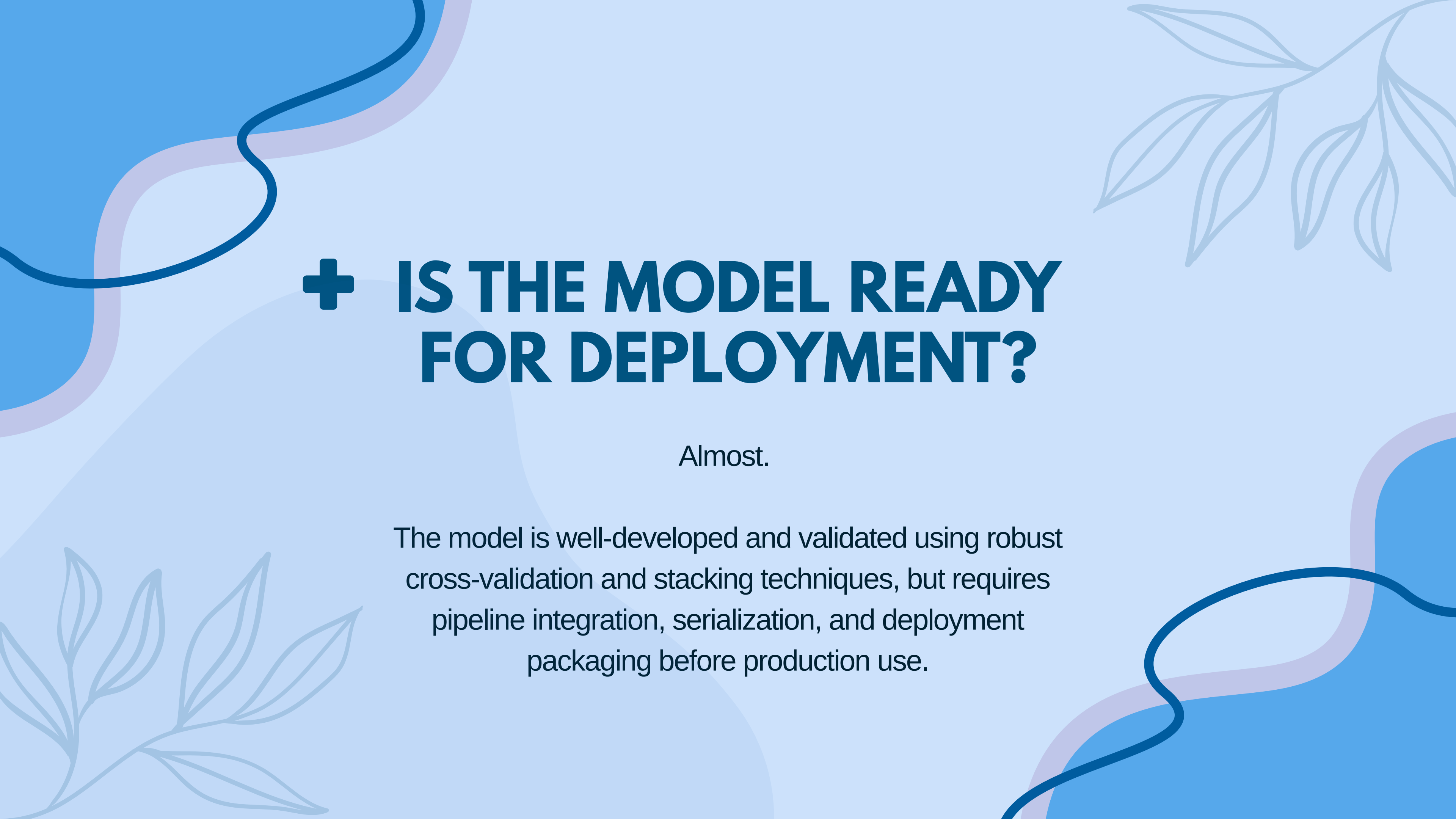
Error Type	Count
Correct	416
FP (False Positive)	65
FN (False Negative)	23
Total	504



LIMITATIONS

WHY DOES IT SEEM TO BE LIKE WE ARE STUCK IN A SPECIFIC RANGE?

- Small dataset → our test sample around ~500 and 50 of them are class 1's (severe imbalance)
- Other papers are including pre-calculated prediction values in their ensemble models but ours is different because we want it to be deployable



+ IS THE MODEL READY FOR DEPLOYMENT?

Almost.

The model is well-developed and validated using robust cross-validation and stacking techniques, but requires pipeline integration, serialization, and deployment packaging before production use.

THANK YOU.

TRY OUT:

- **NESTED**
- **DL**
- **BIAS CHECKS**
- **CHECK XAI TO SEE IF MAKES SENSE**
- **LOOK AT GAPS BETWEEN DEMOGRAPHIC VARIABLES**